Free pythagorean theorem worksheet worksheet 10

Maths | 9th grade | Mixed Operations

1. If x = 3 and y = 5, what is the value of 3x^2 - 4y?

A) 11
B) 16
C) 23
D) 19

2. What is the result of (2^3 * 5) - (18 / 3)?

A) 38
B) 30
C) 32
D) 24

3. Simplify the expression: 4(a + 3) - 2(2a - 5).

A) 2a + 22
B) 2a + 10
C) 4a + 4
D) 0

4. What is the solution to the equation 5(x - 2) + 4 = 3x + 6?

A) x = 4
B) x = 5
C) x = 6
D) x = 7

5. Evaluate: (6 + 9) / 3 + (4 * 2) - sqrt(16).

A) 9
B) 10
C) 11
D) 8

6. If f(x) = 2x^2 - 3x + 1, find f(3).

A) 10
B) 19
C) 16
D) 13

7. Solve for y: 3y/2 + 4 = 10.

A) y = 6
B) y = 4
C) y = 8
D) y = 10

8. What is the value of the expression 7 - 2^3 + 4 * 2?

A) 5
B) 6

C) 7
D) 8

9. Simplify: (x^2 - 4)/(x + 2) when x "` -2.

A) x - 2
B) x + 2
C) x
D) 0

10. Calculate the sum of the arithmetic series: 2 + 5 + 8 + ... + 47.

A) 392
B) 400
C) 425
D) 450

Answer Key

1. D
Substitute x and y into the expression: 3(3)^2 - 4(5) = 27 - 20 = 7.

2. C
Calculate step-by-step: (8 * 5) - 6 = 40 - 6 = 34.

3. B
Expand and combine like terms: 4a + 12 - 4a + 10 = 22.

4. A
Expand the equation: 5x - 10 + 4 = 3x + 6. Simplify to get 2x = 12, so x = 6.

5. A
Perform operations: 15/3 + 8 - 4 = 5 + 8 - 4 = 9.

6. C
Substitute x = 3: 2(3)^2 - 3(3) + 1 = 18 - 9 + 1 = 10.

7. A
Subtract 4 from both sides: 3y/2 = 6. Multiply by 2/3 to isolate y: y = 6.

8. B
Evaluate the expression: 7 - 8 + 8 = 7.

9. A
Factor and simplify: (x + 2)(x - 2)/(x + 2) = x - 2.

10. B
First term a = 2, last term l = 47, common difference d = 3. n = (47 - 2)/3 + 1 = 16. Sum = n/2(a + l) = 16/2(2 + 47) = 400.